# Personal Statement

What if revolutionary technology could transform every facet of life?What started as community service at the Cancer Society led Mahder to a deeper understanding of how small actions can make a big difference. She saw how technology that addresses real health needs can make care more accessible and improve lives for all. Mahder sees technology as a way to solve everyday problems and improve the quality of life for more people.

# Projects

**MyHaki**

**Github link**

MyHaki, an  AI agent-based system was developed after observing the lack of free legal aid service provision in Nairobi, Kenya, during the Gen-Z demonstrations. It connects lawyers with pretrial detainees seeking legal aid services while enabling the Law Society of Kenya to monitor lawyers handling these cases, and offering rewards to motivate them, thereby enhancing access to justice.

- Collected legal documents in various formats, including CSV, PDF, and JSON, and used Pandas to clean and organize them for machine learning. To expand the dataset, I employed Web scraping to gather additional legal documents and information about lawyers.
- Used LangChain's RecursiveCharacterTextSplitter to break down legal text into manageable segments, improving information retrieval. After evaluating different embedding models, chose Legal-BERT for its consistent performance, as DistilBERT struggled with crashes and slower processing.
- Developed a RAG pipeline using  Legal-BERT to provide tailored legal recommendations based on case context. Generated text embeddings with Sentence Transformers and stored them in Supabase with pgvector for efficient, scalable searches to support case prioritization and classification.

**Movie Recommendation System**

**Github link**

Developed a semantic movie recommendation system that gives personalized suggestions from IMDB dataset, matching to what users love for an enjoyable experience.

- Built a movie recommendation system in Google Colab using the IMDB top 1000 dataset to suggest films based on user preferences.
- Loaded the dataset and cleaned it using Pandas, merging movie details into a single MetaText column for streamlined processing. I also lemmatized words with Spacy to standardize the text, making it more consistent for analysis.
- Preprocessed text by joining genres and creating a unified text format for each movie to enable semantic analysis and split movie data into chunks using RecursiveCharacterTextSplitter with a chunk size of 512 tokens to optimize text processing
- Generated embeddings for movie descriptions using the intfloat/ multilingual-e5-small model and stored them in a FAISS vector store for similarity searches and created a RetrievalQA chain with a custom prompt and TinyLlama to generate explanations for recommendations based on genre, director, stars, and plot.

**Tumor Detection with BUSI Dataset**

**Github link**

Developed a machine learning system to analyze and classify breast ultrasound images from the BUSI dataset (sourced from Kaggle) as either tumorous or non-tumorous.

- Analyzed breast ultrasound images from the BUSI dataset from Kaggle in Google Colab to classify images as tumorous and non-tumorous.
- Converted grayscale images to RGB format for compatibility with the model, ensuring consistent input with three color channels.
- Evaluated the model on a test dataset, measuring test loss and accuracy to assess how well it detects tumors.
- Handled image data carefully to ensure proper preprocessing and classification for reliable tumor detection results.

# Mahder Belete

## AI/ML Engineer

## Contact Me


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## Education

**AkiraChix**

codeHive
Diploma in Information Technology
February 2025 – present

**Relevant Courses**

- Languages: Python, SQL
- Libraries & Frameworks
- Sentence Transformers , Tiny-Lama, Pandas, NumPy, scikit-learn, TensorFlow, (all-mpnet-base-v2), legalBERT, LangChain, FAISS, NLTK, SpaCy, Matplotlib, Seaborn, Plotly
- Databases:  PostgreSQL ,ChromaDB ,pgvector, Supabase
- Tools & Platforms
- Jupyter Notebook, Google Colab, GitHub, Python zipfile,FastAPI, BeautifulSoup, JSON utilities,
- Cloud & DevOps: Docker(Docker Hub), GHCR, Google Cloud Run, GitHub Actions (CI/CD)

## References


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